

Victor Pădurean

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Profile

Ph.D. researcher at the Max Planck Institute for Software Systems working on large language models, personalization, and AI for education, with a focus on building and evaluating practical AI systems.

Research Interests: GenAI, RAG, Multimodal AI, Personalization, Human-AI Interaction, AI Evaluation, AI for Education

Education

Ph.D. Max Planck Institute for Software Systems / Saarland University, Germany Sep 2021 - Present
Computer Science

- **Supervisor:** Prof. Adish Singla
- **CS@max planck** program

B.Sc. Technical University of Cluj-Napoca, Romania Oct 2017 - Jun 2021
Computer Science

- **GPA:** 10.0/10.0
- **Thesis:** Thermal Flexibility Estimation for Server-Heaters
- **Advisor:** Dr. Marcel Antal

Selected Publications

- [1] **Victor-Alexandru Pădurean**, Parameswaran Kamalaruban, Nachiket Kotalwar, Alkis Gotovos, and Adish Singla. Inference-Time Personalized Alignment with a Few User Preference Queries. In *NeurIPS*, 2025.
- [2] **Victor-Alexandru Pădurean**, Alkis Gotovos, Ahana Ghosh, Paul Denny, Juho Leinonen, Andrew Luxton-Reilly, James Prather, and Adish Singla. Interleaving Natural Language Prompting with Code Editing for Solving Programming Tasks with Generative AI Models. To appear in *ITiCSE*, 2026.
- [3] **Victor-Alexandru Pădurean**, Paul Denny, Alkis Gotovos, and Adish Singla. Prompt Programming: A Platform for Dialogue-Based Computational Problem Solving with Generative AI Models. In *ITiCSE*, 2025.
- [4] **Victor-Alexandru Pădurean**, Paul Denny, and Adish Singla. BugSpotter: Automated Generation of Code Debugging Exercises. In *SIGCSE TS*, 2025.
- [5] Ruchit Rawal, **Victor-Alexandru Pădurean**, Sven Apel, Adish Singla, and Mariya Toneva. Hints Help Finding and Fixing Bugs Differently in Python and Text-Based Program Representations. In *ICSE*, 2025.
- [6] **Victor-Alexandru Pădurean**, Tung Phung, Nachiket Kotalwar, Michael Liut, Juho Leinonen, Paul Denny, and Adish Singla. Humanizing Automated Programming Feedback: Fine-Tuning Generative Models with Student-Written Feedback. In *EDM*, 2025.
- [7] **Victor-Alexandru Pădurean** and Adish Singla. Benchmarking Generative Models on Computational Thinking Tests in Elementary Visual Programming. In *NeurIPS Datasets and Benchmarks Track*, 2024.
- [8] **Victor-Alexandru Pădurean**, Georgios Tzannetos, and Adish Singla. Neural Task Synthesis for Visual Programming. *TMLR*, 2024.

Work Experience

Amazon, Research Scientist Intern

Italy

Sep 2025 - Jan 2026

- Designed and implemented a knowledge curation pipeline for agentic systems, integrating graph-based RAG, multimodal document processing, and live retrieval.
- Evaluated retrieval strategies across benchmarks and modalities, helping guide architecture choices for production-oriented deployment.
- Built components using AWS Bedrock, LangChain, LightRAG, and SCIP.

Linnify, Data Scientist

Romania

Oct 2020 - Nov 2022

- Researched machine learning approaches for correlating workplace behavioral data with employee stress indicators.
- Contributed to early-stage data science development and experimentation.

Linnify, Backend Developer Intern

Romania

Jul 2020 - Aug 2020

- Developed and deployed an internal tool for project management and team collaboration in a Scrum-based workflow.
- Worked with Python, Django, PostgreSQL, Docker, Bitbucket Pipelines, and Google Cloud.

Montran Corporation, Full Stack Developer Intern

Romania

Jul 2019 - Aug 2019

- Contributed to the development of a banking web application.
- Worked with Java, Spring, MySQL, MongoDB, and Maven.

Skills

Programming: Python, Java, JavaScript, SQL, C

ML / GenAI: PyTorch, TensorFlow/Keras, Hugging Face Transformers, Retrieval-Augmented Generation, LangChain, LangGraph, LightRAG

Data / Databases: NumPy, Pandas, PostgreSQL, MySQL, MongoDB

Frameworks: Django, React, Spring

Tools / Cloud: Git, Docker, Linux, AWS Bedrock, Google Cloud, $\LaTeX$

Honors/Awards

Valedictorian, Computer Science – English Study Program, Technical University of Cluj-Napoca, 2021

Teaching

Generative AI, Saarland University

Winter Semester 2024/25

Teaching Assistant

Advanced Course on Reinforcement Learning, Saarland University

Winter Semester 2022/23

Teaching Assistant

Data Networks, Saarland University

Summer Semester 2022

Teaching Assistant

Service & Mentorship

Reviewer: NeurIPS 2025, TMLR 2025, ITiCSE 2025, ISSEP 2025

Mentor: Mentorship for Relief Program, MPI-SWS - mentoring Othman Shbeir (2024 - ongoing)

Languages

Romanian: Native

English: Proficient (C2)

French: Intermediate (B1)

German: Beginner (A1/A2)